

```
import math
import string
import sys
```

```
def read_file(filename):
    try:
        with open(filename, "r", encoding="utf-8") as f:
            data = f.read()
            return data
    except IOError as e:
        raise IOError(f"Error opening or reading input file: {filename}") from e
```

```
translation_table = str.maketrans(
    string.punctuation + string.ascii_uppercase,
    " " * len(string.punctuation) + string.ascii_lowercase
)
```

```
def get_words_from_line_list(text):
    text = text.translate(translation_table)
    word_list = text.split()
    return word_list
```

```
def count_frequency(word_list):
    D={}
    for new_word in word_list:
        if new_word in D:
            D[new_word] = 1
    return D
```

```
def word_frequencies_for_file(filename):
    line_list = read_file(filename)
    word_list = get_words_from_line_list(line_list)
    freq_mapping = count_frequency(word_list)

    print(f"File {filename}:")
    print(f"{len(line_list)} characters, ")
    print(f"{len(word_list)} words, ")
    print(f"{len(freq_mapping)} distinct words")

    return freq_mapping
```

```
def dotProduct(D1, D2):
    Sum = 0.0
    for key in D1:
        if key in D2: Sum+= D1[key] * D2[key]
    return Sum
```

```
def vector_angle(D1, D2):
    numerator = dotProduct(D1, D2)
    denominator = math.sqrt(dotProduct(D1, D1) * dotProduct(D2, D2))
    if denominator == 0:
        return math.pi / 2
    return math.acos(numerator / denominator)
```

```
def documentSimilarity(filename_1, filename_2):
    sorted_word_list_1 = word_frequencies_for_file(filename_1)
    sorted_word_list_2 = word_frequencies_for_file(filename_2)
    distance = vector_angle(sorted_word_list_1, sorted_word_list_2)

    print(f"The distance between the documents is: {distance:0.6f} (radians)")
    return distance
```

```
filename1 = "1789-Washington.txt"
filename2 = "alice.txt"
```

```
documentSimilarity(filename1, filename2)
```

```
File 1789-Washington.txt:
8613 characters,
1431 words,
0 distinct words
File alice.txt:
142477 characters,
27662 words,
0 distinct words
The distance between the documents is: 1.570796 (radians)
1.5707963267948966
```

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